King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 208 Major Exam I Term 222

Original exam date: 20 February 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. Let $A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & 5 \\ -3 & 3 & -1 \end{bmatrix}$. If $\operatorname{adj} A = \begin{bmatrix} c & 19 & 10 \\ -11 & a & d \\ * & b & -14 \end{bmatrix}$, then $a+b+c+d =$

(a) -1

(b) -5

(c) -4

(d) -3

(e) -2

2. If $y = ae^{-x} + bx - 1$ is a solution of the initial-value problem $y' = x - y,\ y(0) = 10$, then $a + b =$

(a) 10

(b) 13

(c) 12

(d) 11

(e) 14

3. If y is the general solution of the differential equation $y' + y\cot x = \cos x$, then there exists a constant c such that

(a) $y\sin x + \dfrac{1}{4}\sin 2x = c$

(b) $y\cos x + \dfrac{1}{4}\sin 2x = c$

(c) $y\sin x - \dfrac{1}{2}\cos 2x = c$

(d) $y\cos x - \dfrac{1}{4}\sin 2x = c$

(e) $y\sin x + \dfrac{1}{4}\cos 2x = c$

4. The separable differential equation $\dfrac{dy}{dx} = 6x(y-1)^{2/3}$

(a) has $y = 1$ as a singular solution

(b) has $y = 1$ and $y = 1 + x^6$ as singular solutions

(c) has no singular solutions

(d) has no solution

(e) has $y = 1 + x^6$ as a singular solution

5. If $y = y(x)$ is the solution of the initial-value problem

$$\frac{dy}{dx} = x\sqrt{x^2+9}, \qquad y(-4) = 0,$$

then $y(4) =$

(a) -2

(b) 2

(c) 1

(d) 0

(e) -1

6. If the differential equation $(3x^2y^3 + y^4)\,dx + (kx^3y^2 + y^4 + 4xy^3)\,dy = 0$ is an exact differential equation, then $k =$

(a) 2

(b) 1

(c) 3

(d) 4

(e) 5

7. If y is the general solution of the differential equation $(x + ye^y)\dfrac{dy}{dx} = 1$, then there exists a constant c such that $x =$

(a) $\dfrac{1}{2}e^y y^2 + ce^{-y}$

(b) $e^y y^2 + ce^y$

(c) $\dfrac{1}{2}e^y y^2 + ce^y$

(d) ce^{-y}

(e) $e^y y^2 + ce^{-y}$

8. If we have the linear system

$$\begin{aligned} x - 2y + z &= 2 \\ 2x - y - 4z &= 13 \\ x - y - z &= k, \end{aligned}$$

then the system is inconsistent if $k \neq$

(a) 3

(b) 6

(c) 4

(d) 5

(e) 7

9. If $y = y(x)$ is the general solution of the exact differential equation $(x + \tan^{-1} y)\,dx + \dfrac{x+y}{1+y^2}\,dy = 0$, then there exists a constant c such that

(a) $\dfrac{x^2}{2} + \tan^{-1} y + \dfrac{1}{2}\ln(1+y^2) = c$

(b) $\dfrac{x^2}{2} + x\tan^{-1} y + \dfrac{1}{2}\ln(1+y^2) = c$

(c) $\dfrac{x^2}{2} + x\cot^{-1} y - \dfrac{1}{2}\ln(1+y^2) = c$

(d) $\dfrac{x^2}{2} + x\tan^{-1} y - \dfrac{1}{2}\ln(1+y^2) = c$

(e) $x^2 + x\cot^{-1} y + \dfrac{1}{2}\ln(1+y^2) = c$

10. Which of the following matrices are in a reduced echelon matrix form?

(a) $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 1 & 5 \\ 1 & 0 & -7 \\ 0 & 0 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(e) $\begin{bmatrix} 1 & 0 & -3 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

11. A certain city had a population of 25,000 in 1960 and a population of 50,000 in 1970. Assume that population will continue to grow exponentially at a constant rate. What population can its city plans expect in the year 2000?

(a) 300,000

(b) 600,000

(c) 500,000

(d) 200,000

(e) 400,000

12. If $y(x) = A\cosh 3x + B\sinh 3x$ is a solution of the initial-value problem

$$y'' - 9y = 0, \quad y(0) = 5, \quad y'(0) = 12,$$

then $A + B =$

(a) 8

(b) 9

(c) 7

(d) 10

(e) 11

13. If y is the general solution of the differential equation $x^2y' = 1 - x^2 + y^2 - x^2y^2$, then there exists a constant c such that

(a) $\tan^{-1} y + x + 1 = c$

(b) $\cot^{-1} y + x + \dfrac{1}{x} = c$

(c) $\tan y + x + \dfrac{1}{x} = c$

(d) $\tan^{-1} y - x - \dfrac{1}{x} = c$

(e) $\tan^{-1} y + x + \dfrac{1}{x} = c$

14. Let $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $AB = I = BA$ where I is the identity matrix, then $a + b + c + d =$

(a) 2

(b) 1

(c) 0

(d) -2

(e) -1

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$c=-17$, $a=14$, $d=15$, $b=-15$, so the sum is -3.
2	**(c)**	$y'=x-y$ forces $b=1$; $y(0)=a-1=10$ gives $a=11$, so $a+b=12$.
3	**(e)**	$\rho=\sin x$: $(y\sin x)'=\frac{1}{2}\sin 2x$, so $y\sin x+\frac{1}{4}\cos 2x=c$.
4	**(a)**	General solution $y=1+(x^2+c)^3$ contains $1+x^6$ $(c=0)$ but not $y=1$.
5	**(d)**	$y=\frac{1}{3}(x^2+9)^{3/2}+C$ is even and $y(-4)=0$, so $y(4)=0$.
6	**(c)**	$M_y=9x^2y^2+4y^3$, $N_x=3kx^2y^2+4y^3$, so $k=3$.
7	**(c)**	Treat $x(y)$: $x'-x=ye^y$, $\rho=e^{-y}$, so $x=\frac{1}{2}y^2e^y+ce^y$.
8	**(d)**	$R_2-2R_1\Rightarrow y-2z=3$; $R_3-R_1\Rightarrow y-2z=k-2$; consistent iff $k=5$.
9	**(b)**	$F=\frac{x^2}{2}+x\tan^{-1}y+g(y)$, $g'=\frac{y}{1+y^2}$, $g=\frac{1}{2}\ln(1+y^2)$.
10	**(e)**	Only the first matrix has all pivots 1, in staircase order, as lone entries in their columns.
11	**(e)**	$e^{10k}=2$, so $P(40)=25000\cdot 2^4=400{,}000$.
12	**(b)**	$y(0)=A=5$, $y'(0)=3B=12$ so $B=4$ and $A+B=9$.
13	**(e)**	$x^2y'=(1-x^2)(1+y^2)$: $\tan^{-1}y=-\frac{1}{x}-x+c$.
14	**(c)**	$B=A^{-1}=\begin{bmatrix}2 & -1\\ -3 & 2\end{bmatrix}$, entries sum to 0.